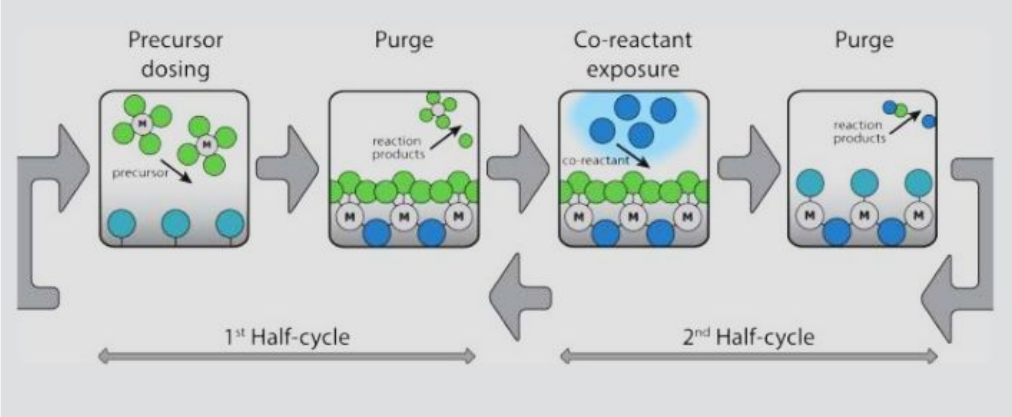
ALD Process & Commercial TCAD Ecosystem

- ALD builds thin films via self-limiting gas-surface reactions.
- Target: **Al2O3** deposition inside a 42 nm pitch FinFET trench.
- **Challenge**: Deep geometry restricts gas flow, pushing precursor transport into the free-molecular Knudsen diffusion regime.



XP4 / XP8 Pulsar (ASMI)	• Deposits ultra-thin high-k gate dielectrics (like hafnium oxide) with atomic precision • Optimized for high-volume manufacturing of advanced transistors (GAA and FinFETs)
STRIKER ALD (Lam Research)	• Deposits highly conformal insulating films (spacers and liners) on single wafers • Crucial for patterning and structural support in advanced logic and memory nodes
Trias e+ EX-II TiN (TEL)	• Deposits titanium nitride barrier layers using high-speed atomic layer deposition (ASFD) • Optimized for single-wafer precision in manufacturing sub-7nm transistor contacts
Fiji G2 PEALD (Veeco)	• Deposits high-quality oxides and nitrides using thermal and plasma-enhanced ALD • Primary research usage in academic cleanrooms, including Georgia Tech's IMS cleanroom
FlexAL ALD (Oxford)	• Provides highly flexible thermal and plasma ALD processing for a wide range of materials • Optimized for research labs and pilot production lines
ALD-150LX (Kurt J. Lesker)	• Enables research-scale thermal and plasma ALD • Commonly integrated into glovebox systems for specialized materials research and prototyping

Baseline Physics Solvers & Geometry Generation

THE PHASE 1 WORKFLOW

- **1. Chemistry:** Cantera solves surface reaction kinetics (**150°C–300°C**).
- **2. Topography:** ViennaPS tracks 3D topography via continuous Level-Set Method. Applies Cantera's Growth Per Cycle (GPC) as the normal advection velocity.

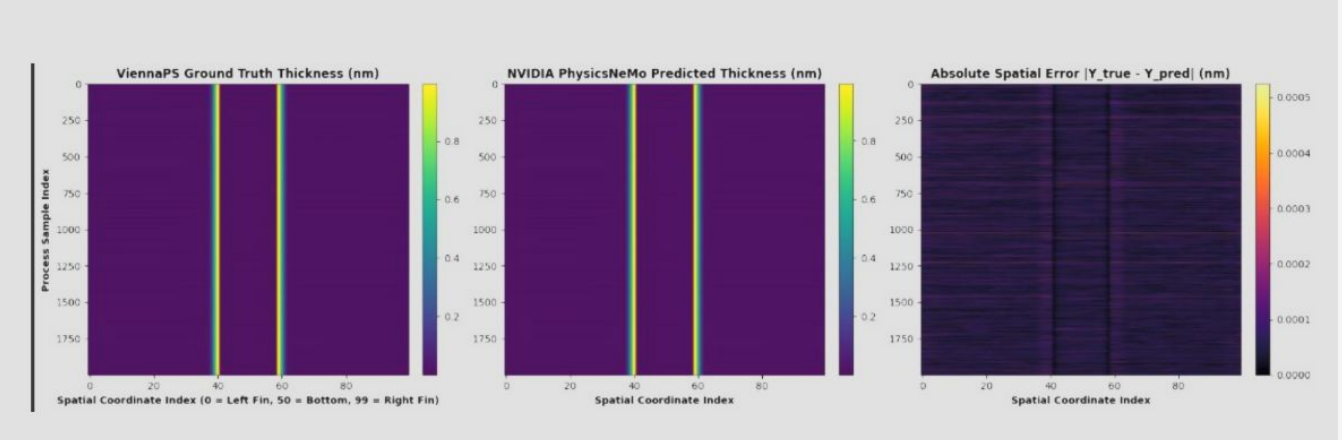
GENERATED BASELINE RESULTS

- Ran **10,000** continuous-cycle simulations.
- Converted 3D .vtp outputs into discrete 1D thickness arrays.
- Mapped film thickness across 100 specific spatial coordinates:
  - **Fin peaks:** Points 0-5, 94-99
  - **Sidewalls:** Points 6-39, 61-93
  - **Trench bottom:** Points 40-60 (heavily shadowed)

AI Acceleration & Factorial Ablation Matrix

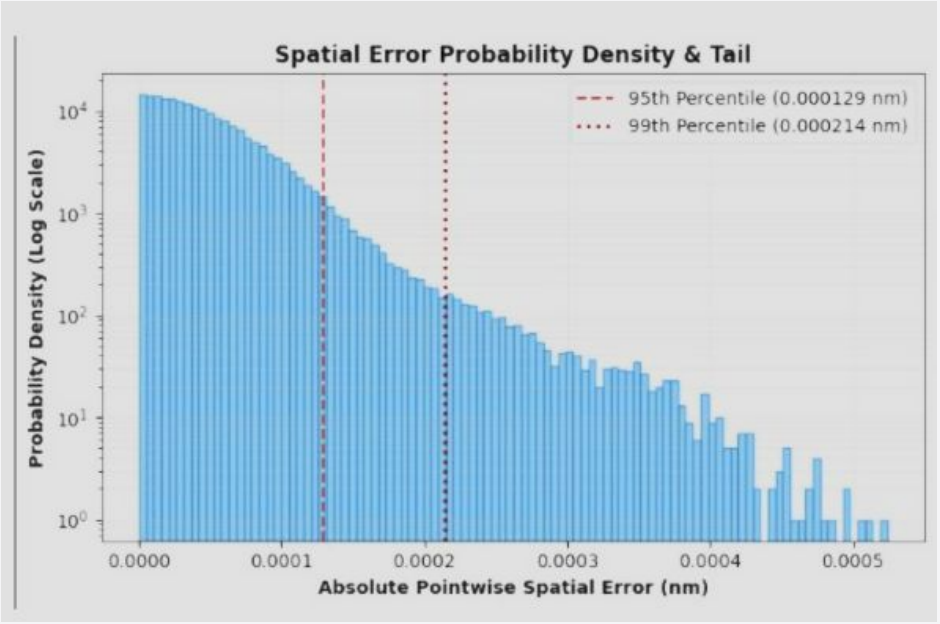
SURROGATE ARCHITECTURE

- Replaced TCAD solver with NVIDIA PhysicsNeMo (FullyConnected, 6 layers, 128 neurons).
- Profile evaluation latency dropped from minutes to **<0.001 ms** per sample.



THE FACTORIAL EXPERIMENT

- Tested 4 data volumes (**N=20 to 8,000**) vs. 4 physics complexity levels (**P0 to P3**).
- Evaluated against a locked 2,000-sample holdout test set to prevent leakage.



PERFORMANCE OUTCOMES

- Unconstrained models diverged on step-coverage at low data volumes.
- **P3 PINN** applied 3.0× spatial loss weighting to the trench bottom, stabilizing conformality.
- Peak model achieved a global Mean Absolute Error of **0.000041 nm** and R^2 of 0.999988

Dataset size (right) Constraints (down)	20	100	1,000	8,000
P0 (no physics)	P0, n = 20	P0, n = 100	P0, n = 1000	P0, n = 8000
P1 (boundary)	P1, n = 20	P1, n = 100	P1, n = 1000	P1, n = 8000
P2 (kinetics)	P2, n = 20	P2, n = 100	P2, n = 1000	P2, n = 8000
P3 (multi-scale)	P3, n = 20	P3, n = 100	P3, n = 1000	P3, n = 8000

Fall Roadmap: Stochastic Topography

THE PHYSICAL FLAW IN PHASE 1

- Continuum solvers dampen high-frequency spatial gradients. The **0.000041 nm** error lacks real-world surface roughness.

THE TPMC-KMC METHODOLOGY

- Replace continuous advection with Ballistic Transport & Kinetic Monte Carlo (kMC).
- **Feature Transport:** ViennaPS computes exact Knudsen gas fluxes inside FinFET walls.
- **Surface Growth:** Python kMC solver executes discrete atomic state transitions based on energy barriers.

EXPECTED OUTCOMES

- **Stochastic simulation = more realistic ALD. Roughness behavior can emerge naturally**
- Retrain P3 surrogate using a **log10** spatial penalty to evaluate scale-agnostic trench defects.

